

Problem 3.7

Given $A(5) = 2500$ and $i = 0.05$.

- (a) What is $A(7)$ assuming simple interest (see Problem 1.5)?
- (b) What is $a(10)$?

Solution

Under simple interest, the amount function is based on

$$A(t) = P(1 + it),$$

and the accumulation function is

$$a(t) = 1 + it.$$

- (a) We first compute the values at times 6 and 7 using simple interest with principal $P = 500$ and $i = 0.05$:

$$A(6) = 500(1 + 0.05 \cdot 6) = 650,$$

and

$$A(7) = 500(1 + 0.05 \cdot 7) = 675.$$

Therefore,

$$i_7 = \frac{A(7) - A(6)}{A(6)} = \frac{675 - 650}{650} = 0.0384615.$$

Also, from the previous year,

$$A(5) = 500(1 + 0.05 \cdot 5) = 625,$$

so

$$i_6 = \frac{A(6) - A(5)}{A(5)} = \frac{650 - 625}{625} = 0.04.$$

Using the procedure shown in the image,

$$A(7) = (1 + i_7)(1 + i_6)A(5).$$

Hence,

$$A(7) = (1.0384615)(1.04)(2500) = 2700.$$

- (b) Since for simple interest

$$a(t) = 1 + it,$$

then

$$a(10) = 1 + (0.05)(10) = 1.5.$$

An easier way to do exercise (a) for the exam is:

$$2500 = P(1 + (0.05)5).$$

So,

$$P = \frac{2500}{1.25} = 2000.$$

Then,

$$A(7) = 2000(1 + (0.05)7) = 2700.$$